

Midterm 1

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Problem 1. Let X and Y be finite sets, suppose $f_1, \dots, f_k$ are a basis of $\text{Hom}_{\text{sets}}(X, \mathbb{C})$ and $h_1, \dots, h_s$ a basis of $\text{Hom}_{\text{sets}}(Y, \mathbb{C})$. Explain why $f_i h_j$ is naturally a basis of $\text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$

Proof. $\forall (x, y) \in X \times Y, \forall i \in [1, \dots, k], \forall j \in [1, \dots, s]$, let $(f_i h_j)(x, y) = f_i(x)h_j(y)$. Thus $\forall i \in [1, \dots, k], \forall j \in [1, \dots, s], f_i h_j \in \text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$.

$\forall c_{i,j} \in \mathbb{C} \ni \sum_{i=1}^k \sum_{j=1}^s c_{i,j} f_i h_j = 0$ Thus $\sum_{i=1}^k \sum_{j=1}^s c_{i,j} f_i h_j = \sum_{i=1}^k f_i \left(\sum_{j=1}^s c_{i,j} h_j \right) = 0$. Since f_i is a basis for $\text{Hom}_{\text{sets}}(X, \mathbb{C})$, it is linearly independent, thus $\sum_{j=1}^s c_{i,j} h_j = 0$. Since $\{h_j\}_j$ is a basis for $\text{Hom}_{\text{sets}}(Y, \mathbb{C})$ and is linearly independent, $c_{i,j} = 0, \forall i, j$. Thus $\{f_i h_j\}_{i,j}$.

Thus $\dim \text{span}(\{f_i h_j\}_{i,j}) = |\{f_i h_j\}_{i,j}| = |X||Y| = |X \times Y|$. Furthermore, since $\text{Hom}_{\text{sets}}(X \times Y, \mathbb{C}) \cong \mathbb{C}^{X \times Y}$, $\dim \text{Hom}_{\text{sets}}(X \times Y, \mathbb{C}) = \dim \mathbb{C}^{X \times Y} = |X \times Y|$. Finally, since $\text{span}(\{f_i h_j\}_{i,j}) \leq \text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$ and $\dim \text{span}(\{f_i h_j\}_{i,j}) = |X \times Y| = \dim \text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$, $\text{span}(\{f_i h_j\}_{i,j}) \cong \text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$. Thus since $\{f_i h_j\}_{i,j}$ are linearly independent and span $\text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$, they are a basis for $\text{Hom}_{\text{sets}}(X \times Y, \mathbb{C})$. $\square$

Problem 2. Let G and H be finite groups, V a rep for G , W a rep for H . Explain how $\text{Hom}(V, W)$ is naturally a rep of $G \times H$ and give a formula for the trace (in terms of tr_V and tr_W).

Proof. Let G, H be finite groups. Let V be a representation for G and let W be a representation for H . Define the representation of $G \times H$ on $\text{Hom}(V, W)$ as follows: $\forall (g, h) \in G \times H, (g, h) \rightarrow L_h \circ R_{g^{-1}}$, where L is the left multiplication operator and R is the right multiplication operator. Thus $\forall f \in \text{Hom}(V, W)$, $(g, h) \cdot f = (L_h \circ R_{g^{-1}})(f) = hf g^{-1}$

$$\forall (g_1, h_1), (g_2, h_2) \in G \times H, \forall f \in \text{Hom}(V, W),$$

$$[(g_1, h_1)(g_2, h_2)] \cdot f = (g_1 g_2, h_1 h_2) \cdot f = h_1 h_2 f g_2^{-1} g_1^{-1} = (g_1, h_1) \cdot (h_2 f g_2^{-1}) = (g_1, h_1) \cdot (g_2, h_2) \cdot f$$

$$\text{tr}_{\text{Hom}(V, W)}((g, h)) = \text{tr}_{\text{Hom}(V, W)}(L_h \circ R_{g^{-1}}). \text{ By claim 2b, } \text{tr}_{\text{Hom}(V, W)}(L_h \circ R_{g^{-1}}) = \text{tr}_W(h) \text{tr}_V(g^{-1}).$$

Since g satisfies the polynomial, $x^{|G|} - 1$, by the linear algebra bootcamp, it is diagonalizable, since the polynomial has distinct roots, with eigenvalues which are solutions to the polynomial and thus part of the $|G|$ -roots of unity. Thus g has eigenvalues $\lambda_1, \dots, \lambda_{\dim V}$ and is conjugate to the

$$\text{matrix } D = \begin{bmatrix} \lambda_1 & 0 & \dots \\ 0 & \ddots & \vdots \\ \vdots & 0 & \lambda_{\dim V} \end{bmatrix} \text{ and } g^{-1} \text{ is conjugate to } D^{-1} = \begin{bmatrix} \lambda_1^{-1} & 0 & \dots \\ 0 & \ddots & \vdots \\ \vdots & 0 & \lambda_{\dim V}^{-1} \end{bmatrix} \text{ and thus has}$$

eigenvalues $\lambda_1^{-1}, \dots, \lambda_{\dim V}^{-1}$. However since all λ s are roots of unity, $\lambda_i^{-1} = \overline{\lambda_i}$. Thus g^{-1} has eigenvalues $\overline{\lambda_1}, \dots, \overline{\lambda_{\dim V}}$. Since the trace is equal to the sum of the eigenvalues, $\text{tr}(g^{-1}) = \overline{\lambda_1} + \dots + \overline{\lambda_{\dim V}} = \overline{\text{tr}(g)}$.

$$\text{Thus } \text{tr}_{\text{Hom}(V, W)}((g, h)) = \text{tr}_W(h) \overline{\text{tr}_V(g)}. \quad \square$$

Claim 2b. $\forall g \in G$ and $\forall h \in H$, $\text{tr}(L_h \circ R_g) = \text{tr}_V(g)\text{tr}_W(h)$. Note this claim is used to prove the problem above.

Proof. $\forall g \in G$, since $g^{|G|} = 1$, g satisfies the polynomial $x^{|G|} - 1 = 0$ which has distinct roots which are a part of $|G|$ th - roots of unity. Thus g is diagonalizable. Similarly, $\forall h \in H$, since $h^{|H|} = 1$, h satisfies the polynomial $x^{|H|} - 1 = 0$ which has distinct roots which are a part of $|G|$ th - roots of unity. Thus h is diagonalizable.

$\forall A \in GL(V)$, $A^T(A^{-1})^T = (A^{-1}A)^T = I$ and $(A^{-1})^T A^T = (AA^{-1})^T = I$. Thus $(A^{-1})^T = (A^T)^{-1}$.

Since g is diagonalizable, $\exists A$, a isomorphism between $C^{\dim V}$ and V , and D , a diagonal transformation, such that $g = ADA^{-1}$. Thus $g^T = (ADA^{-1})^T = (A^{-1})^T D^T A^T = (A^T)^{-1} D A^T$, thus g^T is diagonalizable.

Since h is diagonalizable, $\exists \{X_i\}_i$ a column eigenvector basis of h where the corresponding eigenvalues are $\{a_i\}_i$. Since g^T is diagonalizable, $\exists \{Y_j^T\}_j$ a row eigenvector basis of g where the corresponding eigenvalues are $\{b_j\}_j$.

$$|\{X_i Y_j^T\}_{i,j}| = |\{X_i\}_i| |\{Y_j^T\}_j| = \dim V \dim W = \dim \text{Hom}(V, W).$$

$\forall \alpha_{i,j} \in \mathbb{C} \ni \sum_{i=0}^{\dim W} \sum_{j=1}^{\dim V} \alpha_{i,j} X_i Y_j^T = 0$. $\sum_{i=0}^{\dim W} \sum_{j=1}^{\dim V} \alpha_{i,j} X_i Y_j^T = \sum_{i=0}^{\dim W} X_i \sum_{j=1}^{\dim V} \alpha_{i,j} Y_j^T$. Let $Z_i := \sum_{j=1}^{\dim V} \alpha_{i,j} Y_j^T$ is a row vector. $\sum_{i=1}^{\dim W} X_i Z_i = 0$ is the zero matrix. The only way for the q th column of the matrix to be zero is if, $(\sum_{i=1}^{\dim W} X_i Z_i) e_q = \sum_{i=1}^{\dim W} X_i Z_i e_q = 0$ is the q th basis of W . Since Z_i is a row vector and e_q is a column vector, $Z_i e_q \in \mathbb{C}$ for all q . Thus, $\sum_{i=1}^{\dim W} X_i Z_i e_q = \sum_{i=1}^{\dim W} (Z_i e_q) X_i = 0 \implies Z_i e_q = 0$ since $\{X_i\}_i$ is a basis for W and thus is linearly independent. Since $Z_i e_q = 0$ for all q , it means every component of Z_i is 0, thus $Z_i = 0$. Thus $0 = Z_i = \sum_{j=1}^{\dim V} \alpha_{i,j} Y_j^T$, thus $\alpha_{i,j} = 0$ because $\{Y_j^T\}$ is a basis for V and is linearly independent. Thus $\alpha_{i,j} = 0$ for all i, j , thus $\{X_i Y_j^T\}_{i,j}$ is linearly independent. Thus, since $\text{span}(\{X_i Y_j^T\}_{i,j}) \leq \text{Hom}(V, W)$ and $\dim \text{span}(\{X_i Y_j^T\}_{i,j}) = \dim V \dim W = \dim \text{Hom}(V, W)$, $\text{span}(\{X_i Y_j^T\}_{i,j}) \cong \text{Hom}(V, W)$. Thus since $\{X_i Y_j^T\}_{i,j}$ is linearly independent and spans $\text{Hom}(V, W)$ it is a basis for $\text{Hom}(V, W)$.

$\forall i \in [1, \dots, \dim W], j \in [1, \dots, \dim V]$ $(L_h \circ R_g)(X_i Y_j^T) = h X_i Y_j^T g = a_i X_i b_j Y_j^T = a_i X_i Y_j^T$, thus for all i and j , $X_i Y_j^T$ is an eigenmatrix with an eigenvalue of $a_i b_j$ for the operator, $L_h \circ R_g$.

Since the trace is equal to the sum of the eigenvalues,

$$\text{tr}_{\text{Hom}(V, W)}(L_h \circ R_g) = \sum_{i=1}^{\dim W} \sum_{j=1}^{\dim V} a_i b_j = \left(\sum_{i=1}^{\dim W} a_i \right) \left(\sum_{j=1}^{\dim V} b_j \right) = \text{tr}_W(h) \text{tr}_V(g)$$

□

Problem 3. Show that if V and W (as in 2)) are irreducible then $\text{Hom}(V, W)$ is an irreducible $G \times H$ rep.

Proof. Let G and H be finite groups and let V be an irreducible for G and W an irreducible for H . Let $\langle \cdot, \cdot \rangle$ be a bilinear form on $\text{Hom}_{\text{sets}}(G/G, \mathbb{C})$ such that $\forall f, g \in \text{Hom}_{\text{sets}}(G/G, \mathbb{C})$, $\langle f, g \rangle = \frac{1}{|G|} \sum_{g \in G} f(g) \overline{g(g)}$.

By Orthogonality Relations of Characters, $\langle \text{tr}_v, \text{tr}_v \rangle = 1$ and $\langle \text{tr}_w, \text{tr}_w \rangle = 1$ since V and W are irreducible.

Thus,

$$\begin{aligned}
\langle tr_{Hom(V,W)}, tr_{Hom(V,W)} \rangle &= \frac{1}{|G \times H|} \sum_{(g,h) \in G \times H} tr_{Hom(V,W)}((g,h)) \overline{tr_{Hom(V,W)}((g,h))} \\
&= \frac{1}{|G||H|} \sum_{(g,h) \in G \times H} tr_W(h) \overline{tr_W(h)} tr_V(g) \overline{tr_V(g)} \\
&= \frac{1}{|G||H|} \sum_{(g,h) \in G \times H} tr_W(h) \overline{tr_W(h)} tr_V(g) \overline{tr_V(g)} \\
&= \left(\frac{1}{|G|} \sum_{g \in G} tr_V(g) \overline{tr_V(g)} \right) \left(\frac{1}{|H|} \sum_{h \in H} tr_W(h) \overline{tr_W(h)} \right) \\
&= \langle tr_V, tr_V \rangle \langle tr_W, tr_W \rangle = 1
\end{aligned}$$

Thus by Orthogonality Relations of Characters, since $\langle tr_{Hom(V,W)}, tr_{Hom(V,W)} \rangle = 1$, $Hom(V,W)$ is irreducible. $\square$

Problem 4. Explain why $Hom(V,W)$ over all irreducible reps gives all the irreducible reps of $G \times H$

Proof. Let G and H be finite groups. Let $\{V_i\}_i$ be the set of nonisomorphic irreducible representations of G . Let $\{W_j\}_j$ be the set of nonisomorphic irreducible representations of H . Let $A_{i,j} = Hom(V_i, W_j)$. By theorem that the sum of squares of the dimensions of a group's nonisomorphic irreducibles is equal to the order of the group, $(\sum_i (\dim V_i)^2) = |G|$ and $(\sum_j (\dim W_j)^2) = |H|$

Take the following sum:

$$\begin{aligned}
\sum_{i,j} (\dim A_{i,j})^2 &= \sum_{i,j} \dim V_i^2 \dim W_j^2 \\
&= \left(\sum_i (\dim V_i)^2 \right) \left(\sum_j (\dim W_j)^2 \right) \\
&= |G||H| = |G \times H|
\end{aligned}$$

Since the order of a group is equal to the sum of the squares of the dimensions of its nonisomorphic irreducibles, $|G \times H|$ is equal to the sum of the squares of the dimensions of its nonisomorphic irreducibles. $\sum_{i,j} (\dim A_{i,j})^2$ is a part of the sum which is already equal to $|G \times H|$. Since squares of dimensions are positive, if there were more irreducibles the value of the sum would exceed $|G \times H|$ thus $\sum_{i,j} (\dim A_{i,j})^2$ is the whole sum. Meaning there are no other irreducibles of $G \times H$. Thus $\{Hom(V_i, W_j)\}_{i,j}$ is the set of all nonisomorphic irreducibles of $G \times H$. $\square$

Problem 5. Let $G \times G$ act on G by $(a,b)(g) = agb^{-1}$. This gives an action of $G \times G$ on C^G . Decompose this into irreducible reps of $G \times G$.

Claim 5a. The action of $(a,b) \in G \times G$ on G has a fixed point h if and only if a and b are conjugate by the fixed point, ie $a = C_h(b) = h b h^{-1}$.

Proof. ($\implies$): Suppose the action of $(a, b) \in G \times G$ on G has a fixed point h , then $ahb^{-1} = h \implies a = hbh^{-1}$. Thus a and b are conjugate and they are specifically conjugate by the fixed point, since $a = C_h(b) = hbh^{-1}$.

($\impliedby$): Suppose a and b are conjugate, then $\exists h \in G \ni a = hbh^{-1}$, thus $(a, b)(h) = ahb^{-1} = hbh^{-1}hb^{-1} = h$. Thus the action of (a, b) has a fixed point. $\square$

Claim 5b. *If $g, h \in G$ are fixed points of the action $(a, b) \in G \times G$ then $g, h \in gZ(b)$.*

Proof. Suppose $g, h \in G$ are fixed points of the action of (a, b) on G . Then by claim 1a, $a = gb g^{-1} = hbh^{-1}$. Thus, $b = (h^{-1}g)bg^{-1}h = (h^{-1}g)b(h^{-1}g)^{-1} \implies b(h^{-1}g) = (h^{-1}g)b \implies h^{-1}g \in Z(b)$. Thus $hh^{-1}g = g \in hZ(b)$. However this means, $\exists z \in Z(b) \ni g = hz$ and $hZ(b) = hZ(b) = gZ(b)$. Since $h \in hZ(b)$ and $g \in gZ(b)$, then $g, h \in gZ(b)$. $\square$

Claim 5c. *If $g \in G$ is a fixed point of the action of $(a, b) \in G \times G$ then $gZ(b)$ is the set of all fixed points.*

Proof. Suppose g is a fixed point of the action $(a, b) \in G \times G$. $\forall z \in Z(a)$, $gz \in gZ(b)$, by claim 1a, $a = C_g(b) = gb g^{-1}$. Thus, $(a, b)(gz) = agzb^{-1} = gb g^{-1}gz b^{-1} = gbzb^{-1} = gzb b^{-1}$, since $z \in Z(b)$. Thus $(a, b)(gz) = gzb b^{-1} = gz$. Thus gz is a fixed point. Thus all elements of $gZ(b)$ are fixed points of the action of (a, b) . Furthermore since g is a fixed point, all other fixed points are part of $gZ(b)$ by claim 5b, thus $gZ(b)$ is the set of all the fixed points of the action of (a, b) on G . $\square$

Claim 5. $C^G \cong \bigoplus_{i=1}^{|G/G|} \text{Hom}(V_i, V_i)$

Proof. Let $\{V_i\}$ be the set of all the nonisomorphic irreducible representations of G .

The trace of a permutation representation gives the number of fixed points of the element. $\forall (a, b) \in G \times G$, by claim 5a, $\text{tr}_{\text{reg}}(a, b) = 0$ if a and b are not conjugate since there are no fixed points. By claim 5c, the set of fixed points is given by a coset of $Z(b)$, thus $\text{tr}_{\text{reg}}(a, b) = |Z(b)|$. Let $H_{i,j} = \text{Hom}(V_i, V_j)$. From last semester, $\forall g \in G$, $Z(gbg^{-1}) = gZ(b)g^{-1}$, thus $|Z(b)| = |gZ(b)g^{-1}| = |Z(gbg^{-1})|$.

$\mathbb{C}^G \cong \bigoplus_{i,j} \text{Hom}(V_i, V_j)^{m_{i,j}}$, because every representation is a direct sum of irreducibles.

Thus,

$$m_{i,j} = \langle tr_{reg}, tr_{H_{i,j}} \rangle = \frac{1}{|G \times G|} \sum_{(g,h) \in G \times G} tr_{reg}(g,h) \overline{tr_{H_{i,h}}(g,h)} \quad (1)$$

$$= \frac{1}{|G \times G|} \sum_{[g] \in G/G} \sum_{a,b \in [g]} tr_{reg}(a,b) \overline{tr_{H_{i,h}}(a,b)} \quad (\text{since for nonconjugate elements } tr_{reg} \text{ is } 0) \quad (2)$$

$$= \frac{1}{|G \times G|} \sum_{[g] \in G/G} \sum_{a,b \in [g]} |Z(b)| \overline{tr_{V_j}(b)} tr_{V_i}(a) \quad (3)$$

$$= \frac{1}{|G \times G|} \sum_{[g] \in G/G} \sum_{a,b \in [g]} |Z(b)| \overline{tr_{V_j}(g)} tr_{V_i}(g) \quad (\text{since trace is conjugacy invariant}) \quad (4)$$

$$= \frac{1}{|G \times G|} \sum_{[g] \in G/G} \sum_{a,b \in [g]} |Z(g)| \overline{tr_{V_j}(g)} tr_{V_i}(g) \quad (\text{since } g \text{ is conjugate thus centralizers are conjugate}) \quad (5)$$

$$= \frac{1}{|G \times G|} \sum_{[g] \in G/G} |[g]|^2 Z(g) \overline{tr_{V_j}(g)} tr_{V_i}(g) \quad (6)$$

$$= \frac{1}{|G|^2} \sum_{[g] \in G/G} |G| |[g]| tr_{V_i}(g) \quad (\text{by Orbit Stabilizer}) \quad (7)$$

$$= \frac{1}{|G|} |G| \langle V_i, V_j \rangle = \langle V_i, V_j \rangle = \delta(i,j) \quad (\text{by Orthogonality}). \quad (8)$$

Thus, $\mathbb{C}^G \cong \bigoplus_{i,j} Hom(V_i, V_j)^{\delta(i,j)} = \bigoplus_{i=1}^{|G/G|} Hom(V_i, V_i)$.

So $\mathbb{C}^G \cong \bigoplus_{i=1}^{|G/G|} Hom(V_i, V_i)$. □

Problem 6. Explain our "wierd" formula that $|G|$ is the sum of the squares of the dimensions of the irreducible reps

Proof. Let $\{V_i\}$ be the set of all the nonisomorphic irreducible representations of G .

Since $\dim \mathbb{C}^G = |G|$ and by problem 5, $\mathbb{C}^G \cong \bigoplus_{i=1}^{|G/G|} Hom(V_i, V_i)$, $|G| = \dim \mathbb{C}^G = \dim \left(\bigoplus_{i=1}^{|G/G|} Hom(V_i, V_i) \right)$.

$\dim \mathbb{C}^G = \dim \left(\bigoplus_{i=1}^{|G/G|} Hom(V_i, V_i) \right) = \sum_{i=1}^{|G/G|} \dim Hom(V_i, V_i) = \sum_{i=1}^{|G/G|} \dim V_i \dim V_i = \sum_{i=1}^{|G/G|} \dim V_i^2$.

Thus $|G| = \sum_{i=1}^{|G/G|} \dim V_i^2$. This agrees with the result we got if we decomposed $\mathbb{C}^G$ into irreducibles of G where $\mathbb{C}^G \cong \bigoplus_{i=1}^{|G/G|} V_i^{\dim V_i}$ thus $|G| = \dim \mathbb{C}^G = \sum_{i=1}^{|G/G|} \dim V_i \dim V_i = \sum_{i=1}^{|G/G|} \dim V_i^2$.

Thus our $|G|$ is the sum of squares of the dimensions comes from breaking down $\mathbb{C}^G$ into irreducibles of $G \times G$ or G . □